

Daniel Xie

ds3xie@uwaterloo.ca | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SKILLS

Design & CAD: SolidWorks (CSWA Certified), AutoCAD, Creo Parametric, Ansys, FEA, MATLAB, GD&T, DFM, DFA, FMEA.

Mechanical: Engineering Drafting, Rapid Prototyping, CNC Workflows (Lathe, 5-Axis Mill), Laser Cutting, 3D Printing.

Electronics & Hardware: PCB Design, KiCad, Microcontrollers, Arduino, ESP32, STM32, Sensor Integration, Actuators.

Robotics & Software: C/C++, Python, Java, JavaScript, OpenCV, Computer Vision, MAVLink, Frontend, Git, Linux, MS Office.

EXPERIENCE

Mechatronics Engineering Teaching Assistant

September 2025 – December 2025

Faculty of Engineering – University of Waterloo | Waterloo, Ontario

- Delivered MTE100 course content across labs and tutorials covering **SolidWorks, AutoCAD**, and robotics systems.
- Guided students through **sensor and actuator selection**, including working principles, performance trade-offs, interfacing methods, and integration into embedded control systems.
- Provided guidance on **CAD modeling standards, GD&T tolerancing, and assembly design** for mechatronic systems of 20-50+ components, incorporating **DFM/DFA principles**.
- Provided technical mentorship on **electromechanical integration**, including wiring practices, driver selection, power budgeting, and safety considerations for electromechanical systems.

Undergraduate Research Assistant

May 2025 – August 2025

Prince Polymer Lab – University of Waterloo | Waterloo, Ontario

- Conducted thesis research alongside a Ph.D candidate on biodegradable and recyclable synthetic polymers for **sustainable material applications**.
- Prepared and processed polymer samples using lab techniques such as **casting, curing, and tensile testing**.

Manufacturing Engineering Intern

January 2025 – April 2025

FS Tool Corporation | Toronto, Ontario

- Modelled 50+ multi-purpose metal cutters using **Creo Parametric**, suitable for unique wood-cutting profiles for companies including Mastercraft and Vaughan Tools.
- Maintained outdated company product catalogs by updating **AutoCAD drawings** for over 100 company parts.
- Conducted **tolerance and fit analyses** of products during the machining process to ensure high-quality standards.

PROJECTS

4-DOF High Torque Robot Arm

- Engineered a 4-degree-of-freedom robotic arm in **SolidWorks**, integrating custom-designed **harmonic drive transmission systems** to maximize torque-to-weight ratio.
- Designed geometries for complex components, including **flex spline, circular spline**, and **elliptical wave generator**.
- Analyzed mechanical load limits, gear reduction ratios, and stress requirements for **NEMA 17** stepper motors to ensure stable operation under dynamic loading conditions.

AutoThaw – Intelligent Driveway Salt Dispenser

- Designed and 3D printed a custom enclosure using **SolidWorks and PLA filament**, mounting a DC motor via press interference fit and set screw support with tight tolerances for a zero-slop drivetrain.
- Developed ESP32-based firmware in C++ to poll a **live weather API** and autonomously trigger salt distribution, integrating **Gemini AI** for intelligent dispensing decisions and **ElevenLabs** for live voice alerts.

Automated Antenna Tracking Tower with Waterloo Rocketry

- Engineered a multi-stage automated antenna tower in **SolidWorks**, calculating **torsional loading and overturning moments** to ensure structural stability against variable wind loads.
- Conducted structural analysis using **Ansys** to identify stress concentrations and calculate Factors of Safety for critical load-bearing mounting interfaces.

EDUCATION

University of Waterloo

Bachelor of Applied Science in Honours Mechatronics Engineering | GPA: 93.14%

September 2024 – April 2029 (Expected)

Waterloo, Ontario